

1. Extension: heterogeneous ability

I now consider an extension in which firms have heterogeneous learning abilities. This will shed light on how a firm's credibility correlates with its ability in equilibrium.

The extended model is identical to the model above except for three changes. First, rather than assuming that each firm is endowed with the same ability λ , I assume that each firm i is endowed with an firm-specific ability λ^i , which is common knowledge. Second, for tractability, I restrict attention to a winner-takes-all setting: i.e., I assume $k_n = 0$ for all $n > 1$. Finally, I relax the equilibrium symmetry assumption. Accordingly, I let α^i denote the credibility of firm i .

I obtain an intuitive result: firms with higher ability are more credible in equilibrium.

Proposition 1. *For all (i, j) such that $\lambda^i < \lambda^j$, $\alpha_1^i(p(t)) \leq \alpha_1^j(p(t))$. Furthermore, this inequality is strict whenever $\alpha_1^i(p(t)) < 1$.*

Proposition 1 states that regardless of when a report is made, a firm with higher ability is weakly more credible, and strictly so whenever firms are not fully truthful. Let us consider why this correlation arises. First, note that high ability firms are able to confirm a story more quickly and thus, all else equal, pose a greater preemptive threat in equilibrium. This in turn implies that in comparison to a high-ability firm, a low-ability firm faces a greater preemptive threat. Thus, the low-ability firm finds immediate faking more advantageous. In light of this, the firms' credibilities must adjust in such a way to preserve their respective indifference conditions. This is achieved endogenously by means of a lower credibility for the low-ability firm, which ensures that it has less to gain from faking.

Appendix A Microfoundation for market share

In the main text, I assumed that the the firm's market share from reporting a story is $k_n\alpha$. Here I provide a microfoundation for this.

Let $\mathcal{N} \equiv \{1, \dots, N\}$ denote the set of news firms. Suppose there is a mass $K > 0$ of consumers, who are indexed by x . Each consumer x *subscribes* to some subset S_x of the firms. I.e., for all x , $S_x \subseteq \mathcal{N}$. Let S_x denote consumer x 's *subscription set*. Fixing any $S \subseteq \{1, \dots, N\}$, let $m(S)$ denote the mass of consumers x such that $S_x = S$, where $\sum_{S \subseteq \mathcal{N}} m(S) = K$. Assume that the mass of consumers with a given subscription set does not depend on the identity of the firms within that set, but only on the number of firms in the set. Formally, suppose that there exists $\gamma_0, \gamma_1, \dots, \gamma_N \geq 0$ such that

$$m(S) = \gamma_n \text{ if and only if } |S| = n, \text{ where } \sum_{n=1}^N \gamma_n \binom{N}{n} = K.$$

Define i 's *market share* to be the mass of consumers who read the story. We assume that a consumer reads a story if she both *considers* the story, and finds it optimal to read it. To formalize this, let $\hat{S} \subseteq \mathcal{N}$ denote the set of firms who reported before i . A consumer x will *consider* a story if and only if:

1. The firm is in the consumer's subscription set, i.e., $i \in S_x$.
2. The consumer has not previously considered the story. I.e., $j \notin S_x$ for all $j \in \hat{S}$.

The mass of consumers who consider firm i 's story is then given by

$$\sum_{j=1}^{N-n} \binom{N-n}{j} \gamma_{j+1} \equiv k_n,$$

where n is the order of i 's report. Next, suppose consumer x faces a cost c_x of reading a story. Suppose that c_x is i.i.d. across x , that for any x , c_x is uniformly distributed on $[0, 1]$, and that c_x is independent of x 's consideration set. Then x 's payoff from reading a story is $\mathbb{I}[\theta = 1] - c_x$. That is, the consumer will incur a cost c_x from reading the story, and a benefit of 1 only if the story is true. Meanwhile, the consumer's payoff from not reading a story is $\mathbb{I}[\theta = 0]$. Namely, the consumer enjoys a payoff of 1 from refusing to story that is untrue. Assuming consumers maximize expected utility, x will read the story if and only if

$$\alpha + (1 - \alpha)p - c_x \geq (1 - \alpha)(1 - p) \Leftrightarrow c_i \leq \alpha$$

where α is the credibility of i 's story. Thus i 's market share is $k_n\alpha$.

Appendix B Equilibrium credibility

Here, I justify equation (??) by showing that it is the limit of Bayes-consistent beliefs under a discrete approximation of the game presented in ???. To this end, for any $\varepsilon > 0$, let the ε -approximation of the game be identical to the game presented in section (??), except with the following modification: any report made by a firm on $[0, \varepsilon]$ is observed by all other players (including the consumer) at ε . That is, rather than observing t_i , the players observe $\tilde{t}_i$, where

$$\tilde{t}_i \equiv \max\{t_i, \varepsilon\}$$

At any (p, n) that is on-path, let $\alpha_n^\varepsilon(p)$ denote the firm's credibility, i.e., the consumer's belief that $s_i \leq \varepsilon$ given that $\tilde{t}_i = \varepsilon$, under the ε -approximation of the game. Let α_n denote the right-limit of the α_n^ε . Then:

$$\alpha_n(p) \equiv \lim_{\varepsilon \rightarrow 0+} \alpha_n^\varepsilon(p)$$

I now establish that $\alpha_n(p)$ is given by (??) at any (p, n) on-path.

Claim 1. For any (p, n) on-path,

$$\alpha_n(p) = \begin{cases} \frac{\lambda p}{\lambda p + b_n(p)} & \text{if } F_{p,n}(0) = 0 \\ 0 & \text{if } F_{p,n}(0) > 0 \end{cases}$$

Proof. For any $\varepsilon > 0$, it follows from Bayes Rule that

$$\alpha_n^\varepsilon(p) = \frac{p(1 - e^{-\lambda\varepsilon})}{p(1 - e^{-\lambda\varepsilon}) + F_{p,n}(\varepsilon)e^{-\lambda\varepsilon}}.$$

If $F_{p,n}(0) = 0$, it follows from L'Hôpital's Rule that:

$$\lim_{\varepsilon \rightarrow 0+} \alpha_n^\varepsilon(p) = \frac{\lambda p}{\lambda p + b_n(p)}$$

If $F_{p,n}(0) > 0$, it follows from the right-continuity of $F_{p,n}$ that

$$\lim_{\varepsilon \rightarrow 0+} \alpha_n^\varepsilon(p) = \frac{0}{0 + \lim_{\varepsilon \rightarrow 0+} F_{p,n}(\varepsilon)} = 0.$$

□

Appendix C Proofs: comparative statics

Proof of ??. First, we establish part (a). Fix all other parameters and let $0 < \beta < \tilde{\beta}$. Let α and $\tilde{\alpha}$ denote the equilibrium credibility functions under β and $\tilde{\beta}$, respectively. Fix an n and assume

inductively that the proposition holds for $n + 1$ if $n < N$. Note that for any (p, n) and t , $p(t)$ will be the same under β and $\tilde{\beta}$. Thus to show the above claim, it suffices to show that for any p , $\alpha_n(p)$ is weakly increasing in β , and strictly so whenever $\alpha_n(p) < 1$.

We begin by showing that $\alpha_n(p) = 1$ implies that $\tilde{\alpha}_n(p) = 1$. First, consider the case where $n = N$. By ??, $\alpha_N(p) = 1$ implies that $k_N \leq \beta$. Thus, $k_N < \tilde{\beta}$, which by ?? implies that $\tilde{\alpha}_N(p) = 1$. Next, consider the case where $n < N$, and assume $\alpha_n(p) = 1$. By ??, this implies that $k_n < \beta$ and $p \leq p_n^* \equiv \frac{\beta - k_n}{\beta - k_n/n}$. Further note that

$$\tilde{p}_n^* \equiv \frac{\tilde{\beta} - k_n}{\tilde{\beta} - k_n/n} > \frac{\beta - k_n}{\beta - k_n/n} \equiv p_n^*.$$

Thus, $k_n < \tilde{\beta}$ and $p < \tilde{p}_n^*$, which by ?? implies $\tilde{\alpha}_n(p) = 1$.

Now, suppose that $\alpha_n(p) < 1$. We wish to show that $\tilde{\alpha}_n(p) > \alpha_n(p)$. Suppose by contradiction that $\tilde{\alpha}_n(p) \leq \alpha_n(p)$. It follows from ?? that if $k_n > \tilde{\beta}$,

$$\lim_{q \rightarrow 0^+} \alpha_n(q) = \beta/k_n < \tilde{\beta}/k_n = \lim_{q \rightarrow 0^+} \tilde{\alpha}_n(q).$$

Meanwhile, if $k_n \leq \tilde{\beta}$,

$$\lim_{q \rightarrow \tilde{p}_n^* +} \alpha_n(q) < 1 = \lim_{q \rightarrow \tilde{p}_n^* +} \tilde{\alpha}_n(q).$$

To see why the latter must hold, first consider the case where $n = 1$. It follows from ?? that $\tilde{\alpha}_n(q) = 1$ for all q . Meanwhile, it follows again from ?? that $\alpha_N(q)$ is constant in q , and because $\alpha_N(p) < 1$, $\lim_{q \rightarrow \tilde{p}_n^* +} \alpha_N(q) < 1$. In the case where $n < N$, because $p_n^* < \tilde{p}_n^*$, it follows from ?? that $\alpha_n(\tilde{p}_n^*) < 1$.

Thus, we have that both when $k_n > \tilde{\beta}$ and when $k_n \leq \tilde{\beta}$, there exists some $\hat{p} < p$ such that $\tilde{\alpha}_n(\hat{p}) > \alpha_n(\hat{p})$ and $\tilde{\alpha}_n, \alpha_n$ satisfy (??) on $[\hat{p}, p]$, for their respective value of β . Thus, there exists a $q \in [\hat{p}, p]$ such that $\alpha_n(q) = \tilde{\alpha}_n(q)$ and $\alpha'_n(q) \geq \tilde{\alpha}'_n(q)$. It follows from (??) that in order for the above two conditions to hold, it must be that

$$X \equiv (\beta - \tilde{\beta})\left(\frac{1 - \alpha_n(q)}{\alpha_n(q)}\right)(1 - q) + \frac{V_{q^i, n+1} - \tilde{V}_{q^i, n+1}}{\alpha_n(q)} \geq 0. \quad (1)$$

where V and $\tilde{V}$ denote the value functions under β and $\tilde{\beta}$, respectively. First consider the case where $n = N$. Then $V_{q^i, n+1} = \tilde{V}_{q^i, n+1} = 0$, and thus $X < 0$, contradicting (1).

Next, consider the case where $n < N$. First suppose that $\alpha_{n+1}(q^i) = 1$. It follows from the inductive assumption that $\tilde{\alpha}_{n+1}(q^i) = 1$. Thus, by ??, $V_{q^i, n+1} = \frac{k_{n+1}q^i}{N-n} = \tilde{V}_{q^i, n+1}$. Again this implies that $X < 0$, contradicting (1). Now, suppose that $\alpha_{n+1}(q^i) < 1$. It then follows from ?? that

$$V_{q^i, n+1} = V_{q^i, n+1}(\delta_0) = k_{n+1}\alpha_{n+1}(q^i) - \beta(1 - q^i).$$

Furthermore,

$$\tilde{V}_{q^i, n+1} = \tilde{V}_{q^i, n+1}(\delta_0) = k_{n+1}\tilde{\alpha}_{n+1}(q^i) - \tilde{\beta}(1 - q^i).$$

Thus, recalling that $q^i = \alpha_{n+1}(q) + (1 - \alpha_{n+1}(q))q$, we have

$$V_{q^i, n+1} - \tilde{V}_{q^i, n+1} \leq k_{n+1}(\alpha_{n+1}(q^i) - \tilde{\alpha}_{n+1}(q^i)).$$

Substituting this into the above expression for X , we obtain

$$X \leq \frac{k_{n+1}(\alpha_{n+1}(q^i) - \tilde{\alpha}_{n+1}(q^i))}{\alpha_n(q)} < 0.$$

where the strict inequality follows from the inductive assumption that $\alpha_{n+1}(q^i) < \tilde{\alpha}_{n+1}(q^i)$. Again, this is a contradiction of (1).

Next, let us establish part (b). Let $\tilde{\lambda} > \lambda > 0$, and let $\alpha, \tilde{\alpha}$ denote the equilibria under λ and $\tilde{\lambda}$, respectively, fixing all other parameters. We begin by showing that $\tilde{\alpha}_n(p) = \tilde{\alpha}_n(p)$ for any p and n . Fix an n and assume inductively that if $n < N$, $\alpha_{n+1}(p) = \tilde{\alpha}_{n+1}(p)$ for all p on-path. Let $V, \tilde{V}$ denote the value functions under the equilibria associated with λ and $\tilde{\lambda}$, respectively. Note that $V_{p, n+1} = \tilde{V}_{p, n+1}$ for all p on-path. In the case where $n = N$, $V_{p, n+1} = \tilde{V}_{p, n+1} = 0$, and thus this holds trivially. In the case where $n < N$, this follows from the inductive assumption.

By ??, α_n and $\tilde{\alpha}_n$ must both be a solution to (P) at all (p, n) on-path, which does not depend on λ . By ??, the solution to (P) is unique, and thus $\alpha_n(p) = \tilde{\alpha}_n(p)$ at all (p, n) on-path. Now fixing any p and n , let $p(t)$ and $\tilde{p}(t)$ denote the common beliefs under λ and $\tilde{\lambda}$, respectively. It follows from (??) that $p(t) > \tilde{p}(t)$ for all $t > 0$. Thus, because $\alpha_n(p)$ and $\tilde{\alpha}_n(p)$ are both weakly decreasing in p (??), it follows that $\alpha_n(p(t)) \leq \tilde{\alpha}_n(p(t))$. Furthermore, since $\tilde{\alpha}(p)$ is strictly decreasing in p (??) whenever $\alpha_n(p) < 1$ and $k_N > \beta$, it follows that $\alpha_n(p(t)) < \alpha_n(\tilde{p}(t))$ in this case.

Finally, let us establish part (c). Let α and $\tilde{\alpha}$ denote the equilibria under N and $N + 1$ firms, respectively, fixing all other parameters. We begin by showing that for all p , $\alpha_n(p) \geq \tilde{\alpha}_n(p)$, and $\alpha_n(p) > \tilde{\alpha}_n(p)$ when $\alpha_n(p) < 1$. To this end, fix an $n \in \{1, \dots, N\}$ and assume inductively that the claim holds for $n + 1$ whenever $n < N$.

We begin by showing that $\tilde{\alpha}_n(p) = 1$ implies that $\alpha_n(p) = 1$. Suppose that $\tilde{\alpha}_n(p) = 1$. By ??, $\beta > k_n$ and $p < \tilde{p}_n^* \equiv \frac{\beta - k_n}{\beta - k_n / (N + 1 - n)}$. Because $p_n^* \equiv \frac{\beta - k_n}{\beta - k_n / (N - n)} > \tilde{p}_n^*$, it follows from ?? that $\alpha_n(p) = 1$.

Now consider the case where $\tilde{\alpha}_n(p) < 1$. We wish to show that $\tilde{\alpha}_n(p) < \alpha_n(p)$. To this end, we begin by making the following observation:

$$\text{If } \alpha_n \text{ and } \tilde{\alpha}_n \text{ both satisfy (??) at } q, \text{ and } \alpha_n(q) = \tilde{\alpha}_n(q), \text{ then } \alpha'_n(q) > \tilde{\alpha}'_n(q). \quad (2)$$

Let us now establish this. Note first that for α_n and $\tilde{\alpha}_n$ to both satisfy (??) at q , given that $\alpha_n(q) =$

$\tilde{\alpha}_n(q)$, the following must hold:

$$\alpha'_n(q) = \frac{-1}{k_n(1-q)\alpha_n(q)} \frac{N-n}{N-n+1} (k_n\alpha_n(q) - V_{q^i,n+1} - \beta(1-\alpha_n(q))(1-q))$$

$$\tilde{\alpha}'_n(q) = \frac{-1}{k_n(1-q)\alpha_n(q)} \frac{N-n+1}{N-n+2} (k_n\alpha_n(q) - \tilde{V}_{q^i,n+1} - \beta(1-\alpha_n(q))(1-q)),$$

where V and $\tilde{V}$ denote the value functions under the equilibria with N and $N+1$ total firms, respectively. Note that if $n = N$, $\alpha'_n(q) = 0$. Meanwhile, by ??, $\tilde{\alpha}'_n(q) < 0$. Thus, $\tilde{\alpha}'_n(q) < \alpha'_n(q)$ must hold. Next, consider the case where $n < N$. We begin by observing that $V_{q^i,n+1} > \tilde{V}_{q^i,n+1}$. To see why this must hold, first consider the case where $\tilde{\alpha}_{n+1}(q^i) = 1$. It then follows from the inductive assumption that $\alpha_n(q^i) = 1$. Then, by ??,

$$\tilde{V}_{q^i,n+1} = \tilde{V}_{q^i,n+1}(\delta_\infty) = \frac{k_{n+1}q^i}{N-n} < \frac{k_{n+1}q^i}{N-n-1} = V_{q^i,n+1}(\delta_\infty) = V_{q^i,n+1}.$$

Next, consider the case where $\tilde{\alpha}_n(q^i) < 1$. In this case, it follows from ?? that

$$\begin{aligned} \tilde{V}_{q^i,n+1} &= \tilde{V}_{q^i,n+1}(\delta_0) = k_{n+1}\tilde{\alpha}_{n+1}(q^i) - \beta(1-q^i) < k_{n+1}\alpha_{n+1}(q^i) - \beta(1-q^i) \\ &= V_{q^i,n+1}(\delta_0) \leq V_{q^i,n+1}, \end{aligned}$$

where the strict inequality follows from the inductive assumption made above. Examining the two ODEs listed above, since by ??, $\alpha'_n(q) \leq 0$, it follows that $\tilde{\alpha}'_n(q) < \alpha'_n(q)$.

Now, assume by contradiction that $\alpha_n(p) \leq \tilde{\alpha}_n(p)$. We begin by showing that there exists a $q^* < p$ such that $\tilde{\alpha}_n(q^*) < \alpha_n(q^*)$. First consider the case where $k_n \geq \beta$. Then, by ??,

$$\lim_{q \rightarrow 0+} \alpha_n(q) = \lim_{q \rightarrow 0+} \tilde{\alpha}_n(q) = \frac{\beta}{k_n}.$$

Then, by the continuous differentiability of α_n and $\tilde{\alpha}_n$ on $(0, p)$, it follows from Equation 2 that for some $q^* < p$ sufficiently small $\alpha_n(q^*) > \tilde{\alpha}_n(q^*)$. Next, consider the case where $k_n < \beta$, and let $p_n^* \equiv \frac{\beta-k_n}{\beta/(N-n+1)-k_n}$. Note by ?? that $\alpha_n(p_n^*) = 1$. Meanwhile, because $p_n^* < \tilde{p}_n^* \equiv \frac{\beta-k_n}{\beta/(N-n+2)-k_n}$, it follows from ?? that $\tilde{\alpha}_n(p_n^*) < 1$, and thus, we have for $q^* = p_n^*$, $\tilde{\alpha}_n(q^*) < \alpha_n(q^*)$.

Since $\tilde{\alpha}_n(q^*) < \alpha_n(q^*)$ and $\tilde{\alpha}_n(p) \geq \alpha_n(p)$, by the continuous differentiability of α on $[q^*, p]$, there must exist some $q \in (q^*, p]$ such that $\alpha_n(q) = \tilde{\alpha}_n(q)$ and $\alpha'_n(q) \leq \tilde{\alpha}'_n(q)$. However, this is a contradiction of (2).

Now fixing any p and n , let $p(t)$ and $\tilde{p}(t)$ denote the common beliefs under N and $N+1$ firms, respectively. We wish to show that on some interval $[0, \bar{t}]$, where $\bar{t} > 0$, $\alpha_n(p(t)) \geq \tilde{\alpha}_n(\tilde{p}(t))$ is weakly increasing in t , and strictly so whenever $\alpha_n(p(t)) < 1$. First consider the case where $\alpha_n(p(t)) = 1$. In this case, the statement holds trivially. Next, consider the case where $\alpha_n(p) < 1$. It follows from the above that $\alpha_n(p) > \tilde{\alpha}_n(p)$. Now note that it follows from (??) that $\lim_{t \rightarrow 0+} p(t) - \tilde{p}(t) = 0$. Since

$\alpha_n(p(t))$ and $\tilde{\alpha}_n(\tilde{p}(t))$ are both continuous in t (??), it follows that for some $\bar{t} > 0$, $\alpha_n(p(t)) > \tilde{\alpha}_n(\tilde{p}(t))$ for all $t \in [0, \bar{t}]$. $\square$

Appendix D Proofs: heterogenous learning abilities

Here, we consider the extended model presented in [Section 1](#). The objective is to establish [Proposition 1](#). This proof will require extending certain results established in the baseline model to the extended model.

Regarding Lemmas 1-4, I will take for granted that these hold under the extended model. Formal proofs of this are omitted as all proofs presented under the baseline model will apply to the extended setting as well.

Next, I establish that ?? holds under the extended model. This claim is presented as [Proposition 1'](#). In the analysis below, I let $V_{p,n}^i$ denote firm i 's value.

Proposition 1'. *For all s , there exists a $p^{i*} \in (0, 1]$ such that at any p on-path, $\alpha_1^i(p) = 1$ if and only if the following two conditions hold:*

1. $k_1 \leq \beta$
2. $p \leq p^{i*}$

Furthermore, $p^{j*} > p^{i*}$ whenever $\lambda^j > \lambda^i$ and $n > 1$.

Proof. Fix an i . Suppose that $k_1 \leq \beta$. By identical reasoning as ??, for all $q < \frac{\beta - k_1}{k_1}$, $\alpha_1^i(q) = 1$. Let

$$p^{i*} \equiv \sup\{p | \alpha_1^i(p) = 1 \text{ for all } q < p\}.$$

It follows by definition that $\alpha_1^i(p) = 1$ for all $p \leq p^{i*}$.

Next, we will show that $\alpha_1^i(q) < 1$ whenever $k_1 > \beta$ or $p > p^{i*}$. Suppose not by contradiction. First, consider the case where $k_1 > \beta$ and $\alpha_1^i(p) = 1$ for some p . Then we have that

$$V_{p,1}^i(\delta_0) = k_1 p + (k_1 - \beta)(1 - p) > k_1 p \leq V_{p,1}^i(\delta_\infty)$$

Thus, i can profitably deviate at p . Contradiction. Next, consider the case where $q > p^{i*}$ and $\alpha_1^i(p) = 1$. In this case, a contradiction follows from identical reasoning to what is presented in ??.

Finally, we show that $p^{j*} > p^{i*}$ whenever $\lambda^j > \lambda^i$. Suppose by contradiction that $p^{j*} \leq p^{i*}$. Note that because j is truth telling at $(p_S^{j*}, n = 1)$, $V_{p_1^{j*},1}^j(\delta_\infty) \geq V_{p^{j*},1}^j(\delta_0)$. Furthermore, because $p^{j*} \leq p^{i*}$, i is also truthful at $(p_n^{j*}, n = 1)$. Thus,

$$V_{p_1^{j*},1}^j(\delta_0) = V_{p_1^{j*},1}^i(\delta_\infty) = k_1 - \beta(1 - p).$$

Now, note that because $\lambda^j > \lambda^i$,

$$V_{p_1^{j*},1}^j(\delta_\infty) > V_{p_1^{j*},1}^i(\delta_\infty).$$

Combining these inequalities we have $V_{p_1^{j*},1}^i(\delta_\infty) < V_{p_1^{j*},1}^i(\delta_0)$. However, because $\alpha_1^i(p^{j*}) = 1$, $V_{p_n^{j*},1}^j = V_{p_n^{j*},1}^j(\delta_\infty)$. Contradiction. $\square$

Next, we extend ?? to this setting. Note this entails deriving an ODE that applies to this extended model, (ODE').

Proposition 2'. *In equilibrium, for any p on-path, if $k_1 \geq \beta$ or $p > p^{i*}$, then the following must be satisfied:*

$$\alpha_1^{i'}(p) = -\beta - \frac{\sum_{j \neq i} \frac{\lambda^j}{\alpha_1^j(p)}}{\sum_j \lambda^j (1-p)} [\alpha^i(p) - \beta(1-p)]. \quad (\text{ODE}')$$

In addition, $\lim_{p \rightarrow 0+} \alpha_1^i(p) = \beta/k_1$ must hold if $k_1 > \beta$, and $\lim_{p \rightarrow p^{i}+} \alpha_1^i(p) = 1$ if $k_1 \leq \beta$.*

Proof. Let us first establish that (ODE') must hold under the conditions specified.

When $k_1 \geq \beta$ or $p > p^{i*}$, it follows from Proposition 1' that $\alpha_1^i(p(t)) < 1$. It then follows from ?? that there exists an $\varepsilon > 0$ such that for all $\Delta \in (0, \varepsilon)$,

$$\frac{V_{p,1}^i(\delta_\Delta) - V_{p,1}^i(\delta_0)}{\Delta} = 0.$$

Recall that $V_{p,1}^i(\delta_0) = k_1 \alpha_1^i(p) - \beta(1-p)$. Meanwhile,

$$V_{p,1}^i(\delta_\Delta) = \int_0^\Delta k_1 \alpha_1^i(p(s)) \Psi^i(s) ds + (1 - \sum_j \lim_{s \rightarrow \Delta-} \Psi^j(s)) [k_1 \alpha_1^i(p(\Delta)) - \beta(1-p(\Delta))],$$

where Ψ is the first-report distribution associated with the strategy profile in which i plays δ_∞ and all $j \neq i$ play $F_{p,1}$. Specifically, for all $s > 0$,

$$\Psi^i(s) = p \lambda^i \int_0^s e^{-\Pi_{j \in S} \lambda^j r} \prod_{j \neq i} (1 - F_{p,1}^j(r)) dr$$

and for $j \neq i$,

$$\begin{aligned} \Psi^j(s) = & p \int_0^s e^{-\Pi_{k \neq j} \lambda^k r} \prod_{k \neq i \neq j} (1 - F_{p,1}^k(r)) d(-e^{-\lambda^j r} (1 - F_{p,1}^j(r))) \\ & + (1-p) \int_0^s \prod_{k \neq i \neq j} (1 - F_{p,1}^k(r)) dF_{p,1}^j(r). \end{aligned}$$

Substituting these two expressions into the above equation for $V_{p,1}^i(\delta_0)$ and following the same sequence of steps in ?? yields (ODE').

The two limit conditions are established by the same reasoning presented in ??.

□

Proof of Proposition 1. Fix any (i, j) such that $\lambda^i > \lambda^j$. We want to show that $\alpha_1^i(p(t)) \leq \alpha_i^j(p(t))$ and that $\alpha_1^i(p(t)) < \alpha_i^j(p(t))$ whenever $\alpha_1^i(p(t)) < 1$. First suppose $\alpha_1^i(p) = 1$. In this case, $\alpha_1^i(p) \geq \alpha_1^j(p)$ is trivially satisfied.

Next, suppose $\alpha_1^i(p) < 1$. We want to show that $\alpha_1^i(p) > \alpha_1^j(p)$. Suppose by contradiction that $\alpha_1^i(p) \leq \alpha_1^j(p)$. First consider the case where $k_1 < \beta$. Then, let

$$q^* \equiv \inf\{q | \alpha_1^j(p) < 1 \text{ and } \alpha_1^j(p) < \alpha_1^i(p)\}.$$

Because the α_1^i are continuous, it follows from Proposition 1', and the assumption that $\alpha_1^i(p) \leq \alpha_1^j(p)$, that $q^* < p$ exists. Again, by continuity, $\alpha_1^j(q^*) = \alpha_1^i(q^*)$. It then follows from (ODE') that $\alpha_1^{j'}(q^*) < \alpha_1^{i'}(q^*)$. But this implies that for some $q > q^*$, $\alpha_1^j(q) > \alpha_1^i(q)$. Contradiction.

Next, consider the case where $k_1 \geq \beta$. Recall by Proposition 2' that $\lim_{p \rightarrow 0+} \alpha_1^i(p) = \lim_{p \rightarrow 0+} \alpha_1^j(p)$. Thus, there exists some $q \in (0, p]$ such that $\alpha_1^i(p) \leq \alpha_1^j(p)$ and $\alpha_1^{i'}(p) \leq \alpha_1^{j'}(p)$. However, it again follows from (ODE') that $\alpha_1^{i'}(p) > \alpha_1^{j'}(p)$. Contradiction.

□